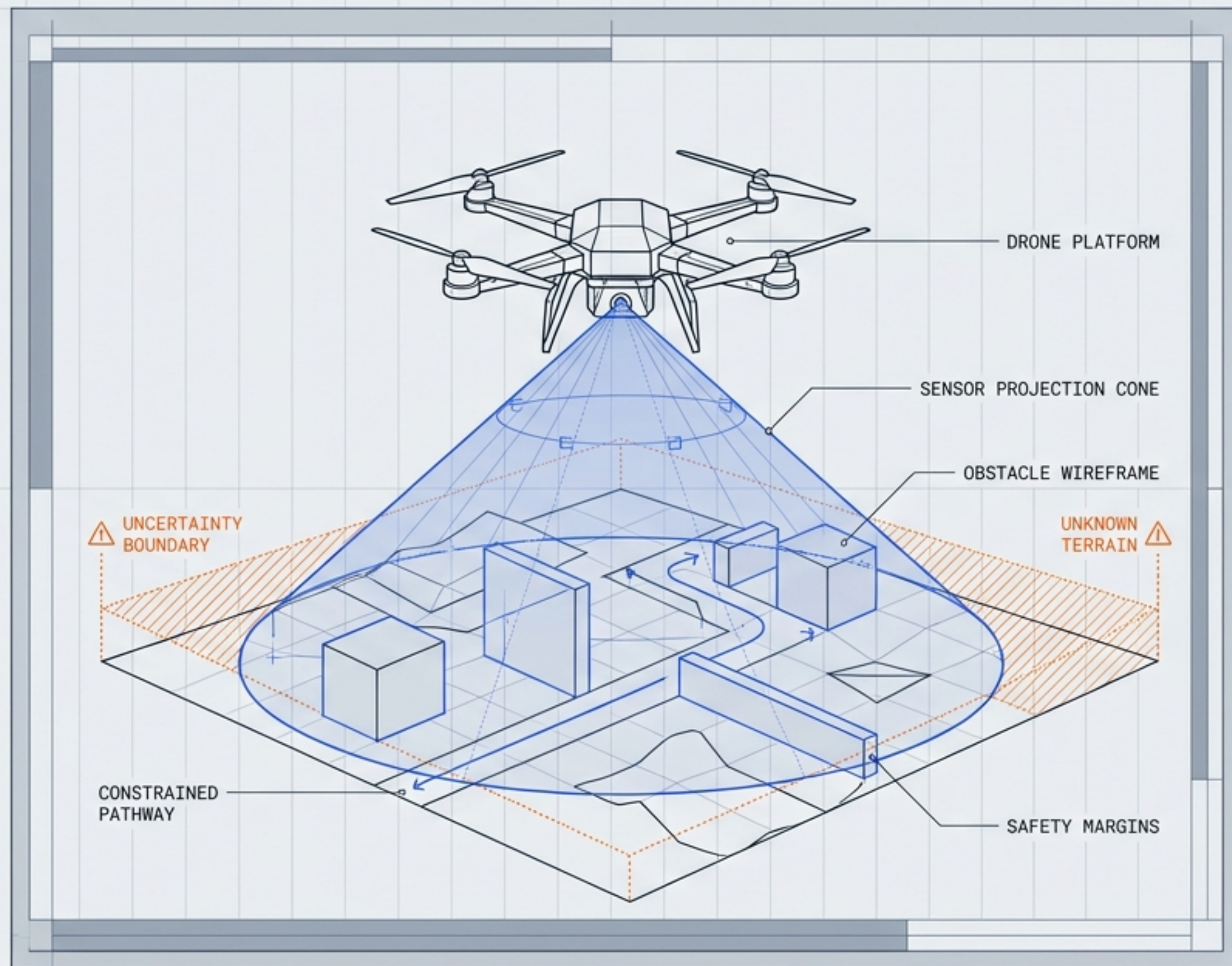
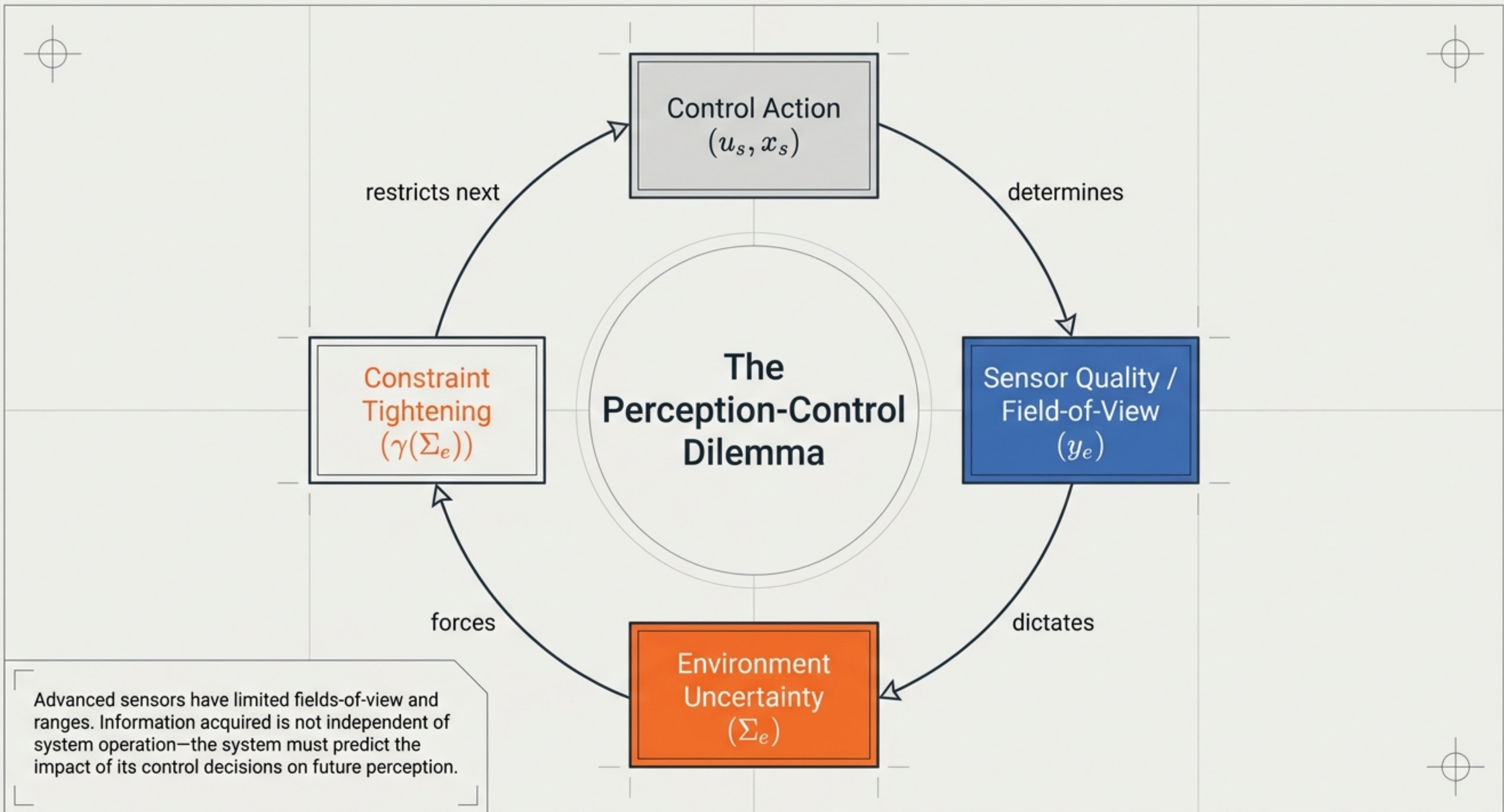


Perception-Aware Model Predictive Control

Navigating Constrained
& Unknown Environments

Based on research by Mitsubishi
Electric Research Laboratories
(MERL) & UC Berkeley
(Bonzanini, Mesbah, Di Cairano)

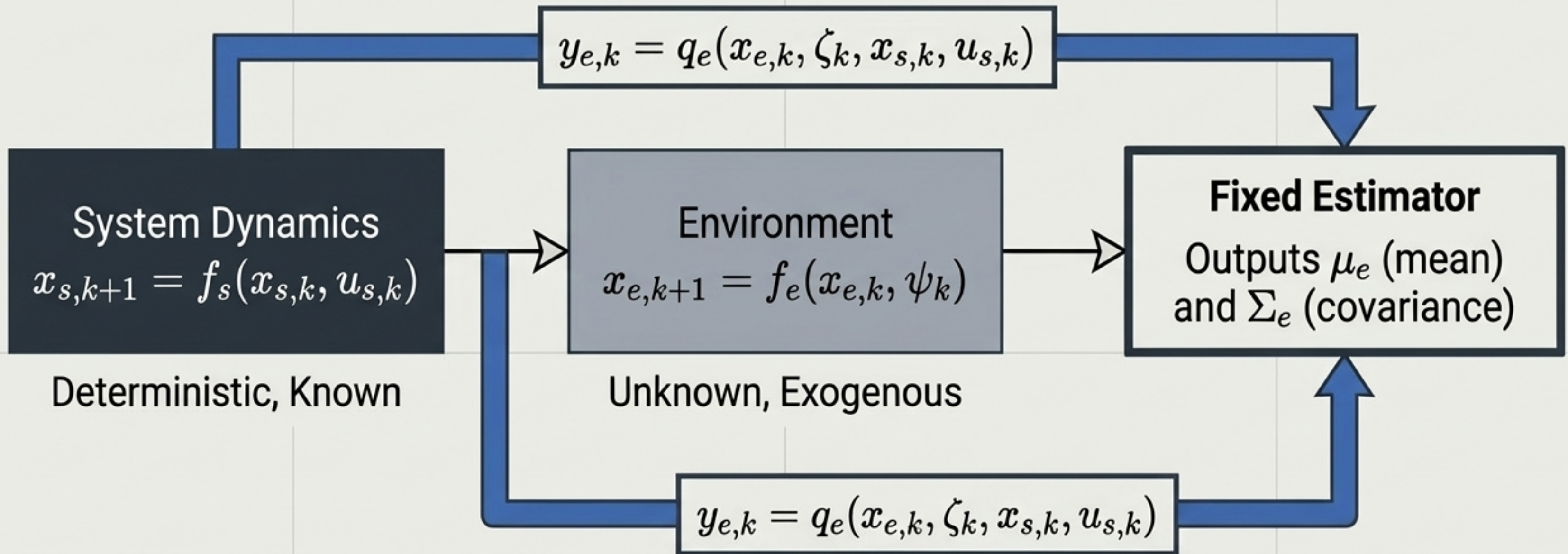




APPROACH	ENVIRONMENT KNOWLEDGE	CONSTRAINT HANDLING	PERCEPTION INTEGRATION
Standard Deterministic MPC	Assumes Known	Rigid	None
Standard Stochastic MPC	Unknown	Probabilistic	Passive (Reacts to uncertainty)
Perception-Aware Chance-Constrained MPC (PAC-MPC)	Unknown	Probabilistic & Dynamic	Active (Predicts and manipulates uncertainty)



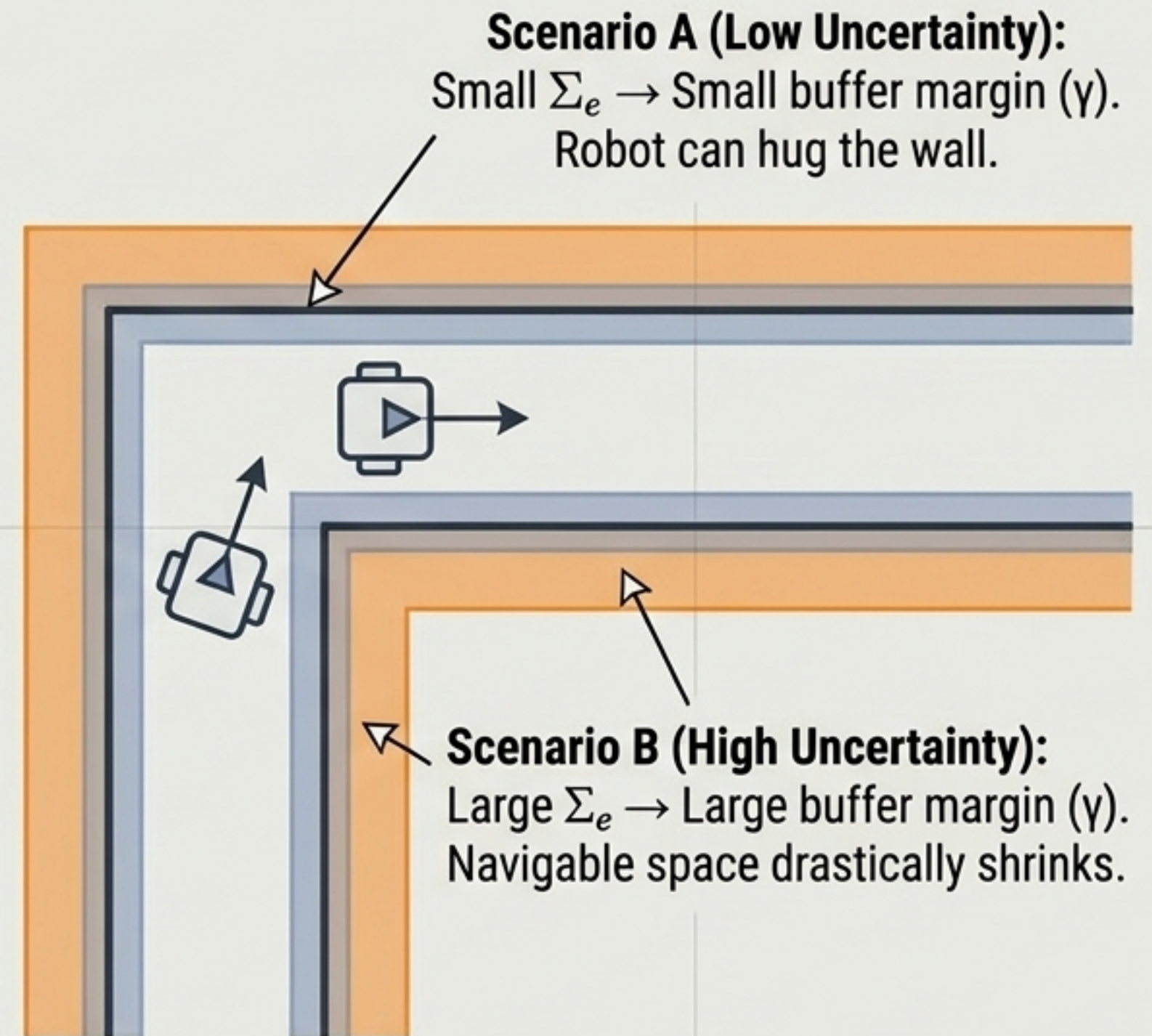
The **PAC-MPC Advantage**: Actively predicting how physical movement alters perception quality, allowing the system to gather information to prevent constraint violation.



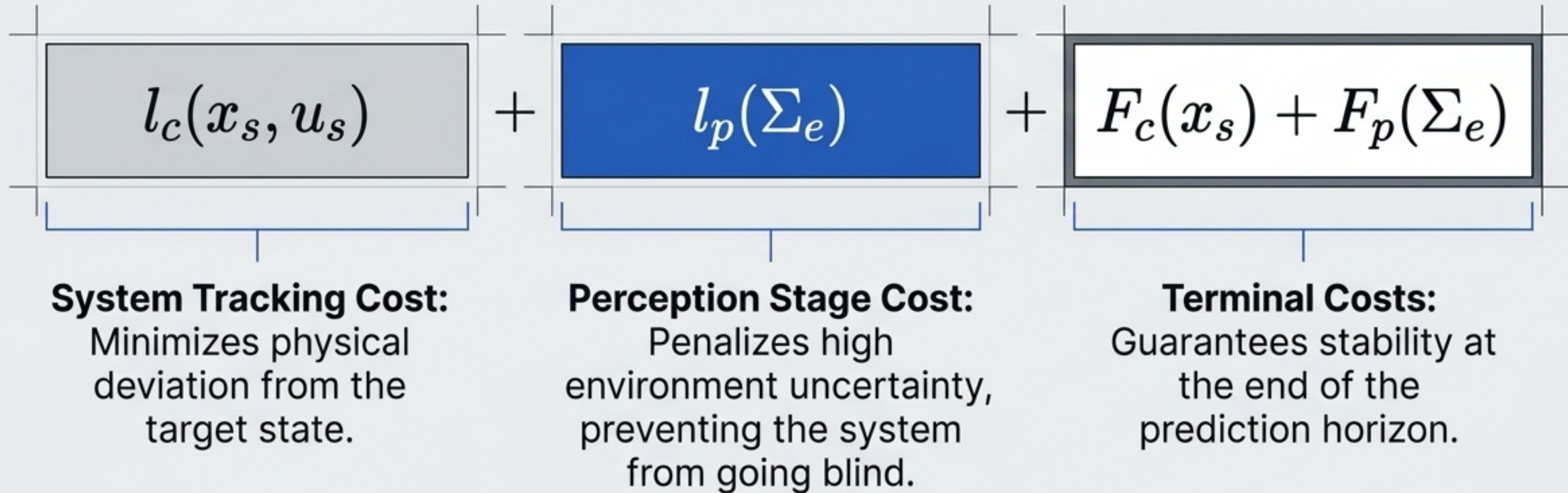
The **System** and **Environment** are decoupled dynamically, but fundamentally linked through perception. The controller cannot redesign the estimator, but it *can* leverage its system-dependent performance.

$$P(h_s x_s + h_e x_e \leq h_b) \geq 1 - \epsilon$$

Enforcing Safety. Uncertainty forces conservative driving. PAC-MPC dynamically tightens constraints based on predicted covariance to guarantee a safe trajectory (e.g., 95% collision avoidance).



$$V_N(x_s, U_k, \Sigma_e, r_k) = [\text{Stage Costs}] + [\text{Terminal Costs}]$$



Unlike standard MPC, PAC-MPC simultaneously minimizes physical tracking error *and* environmental blindness.

Pillar 1: Recursive Feasibility (Theorem 1)

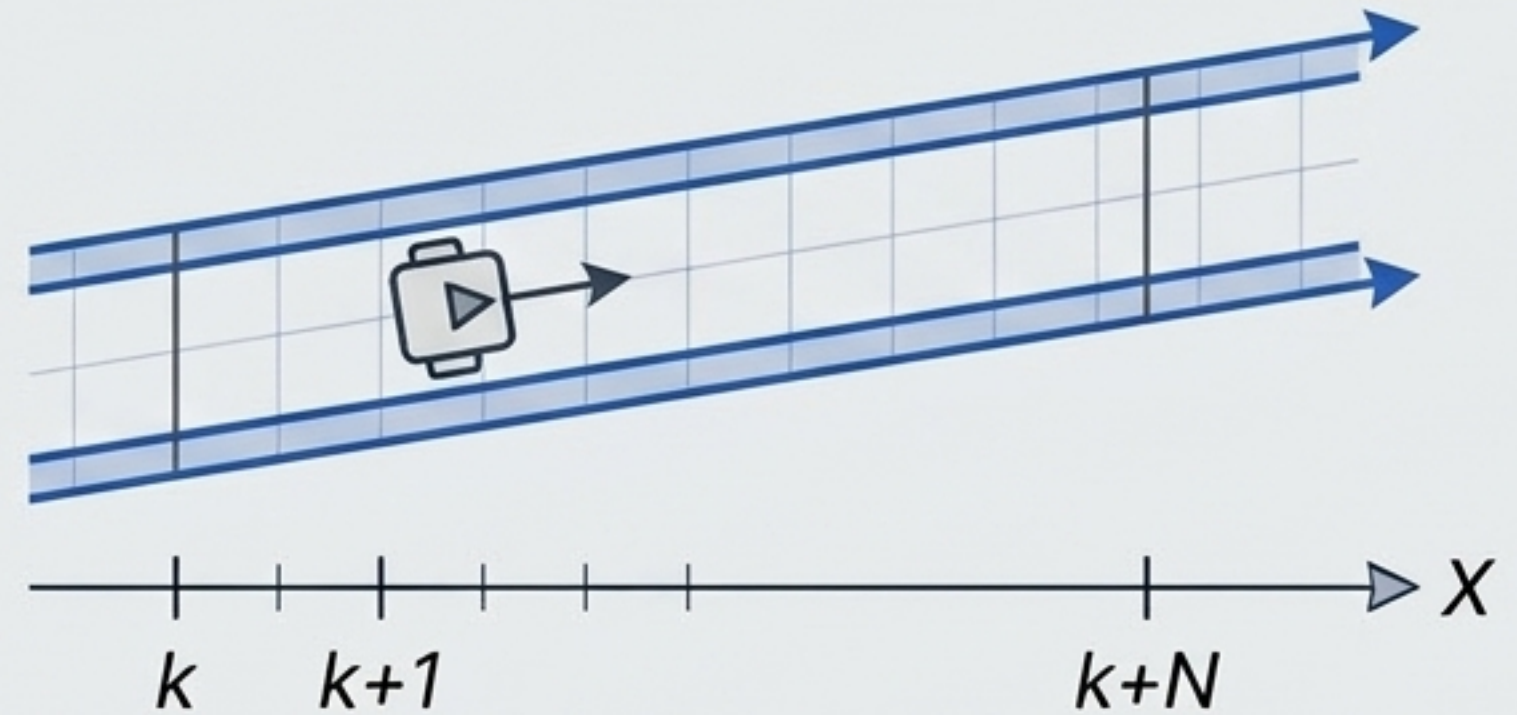
IF the predictor does not underestimate constraint tightening (Assumptions 3 & 4):

$$\gamma(\hat{M}_e) \geq \gamma(M_e)$$

AND

an initial safe state exists

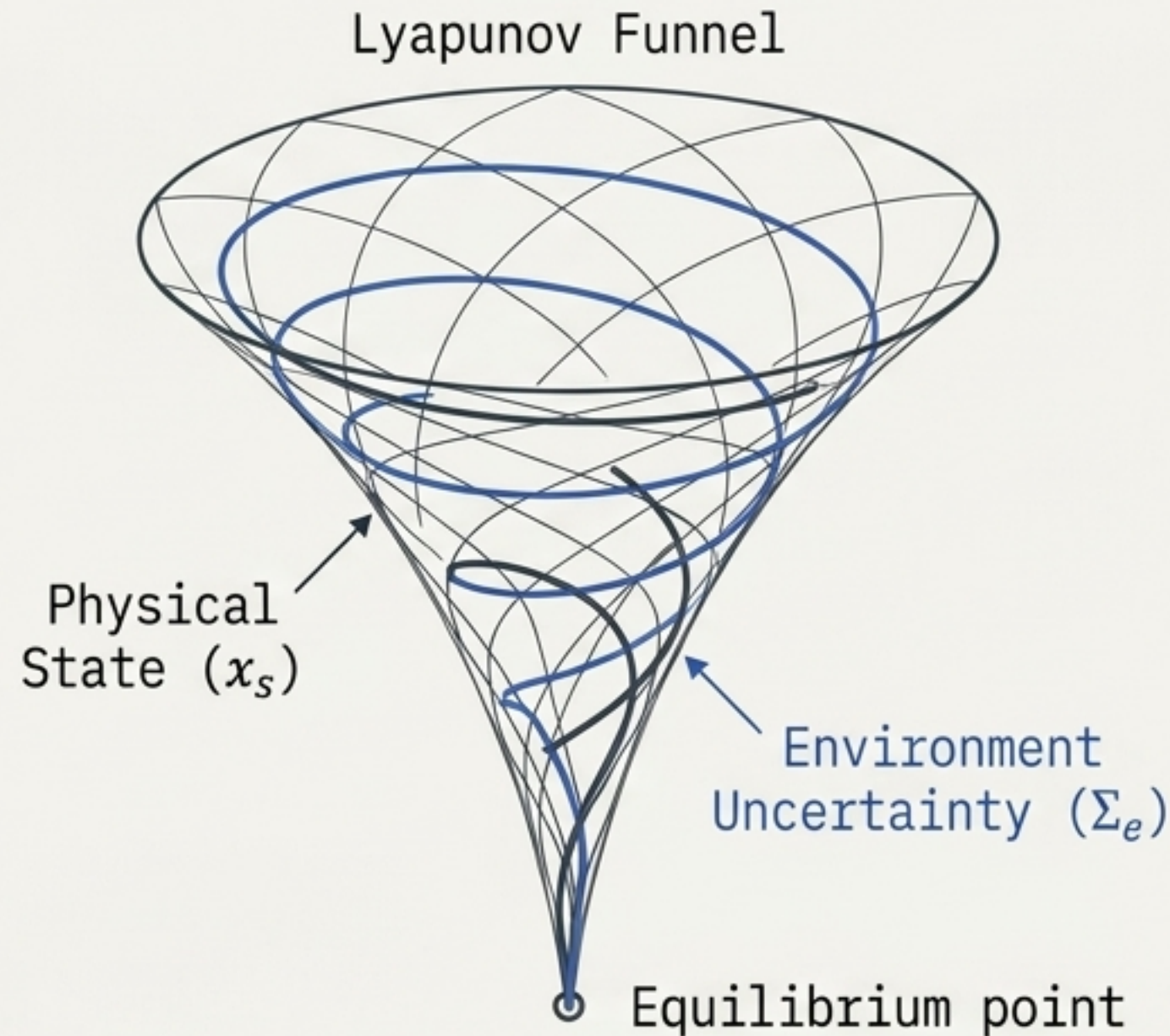
THEN the Terminal Set (Z_f) ensures a mathematically feasible path always exists in the future.



Pillar 2: Asymptotic Stability (Theorem 2)

(Condition):

Requires a terminal control cost that bounds perception uncertainty (Assumptions 5 & 6).

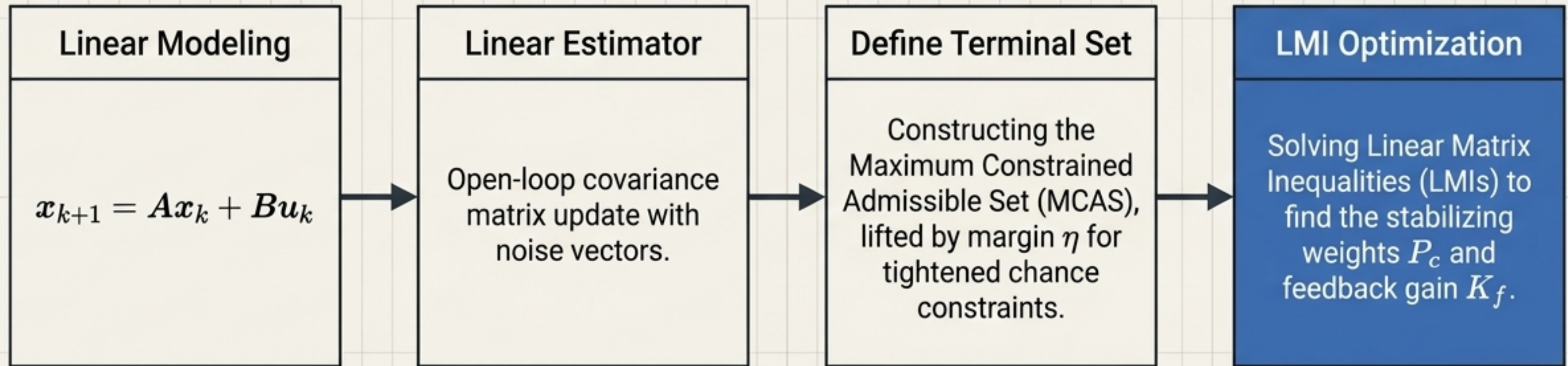


(Guarantee):

Asymptotic Stability of the Augmented State $\xi = (x_s, \Sigma_e)$.

PAC-MPC guarantees that the physical trajectory and the sensory clarity converge simultaneously. The system stabilizes its position while successfully stabilizing its vision.

Constructive Design Pipeline for Linear Dynamics



Constructive Design Pipeline. For linear systems, the dense nonlinear assumptions of PAC-MPC can be systematically resolved offline using standard LMI optimization, enabling real-time deployment.

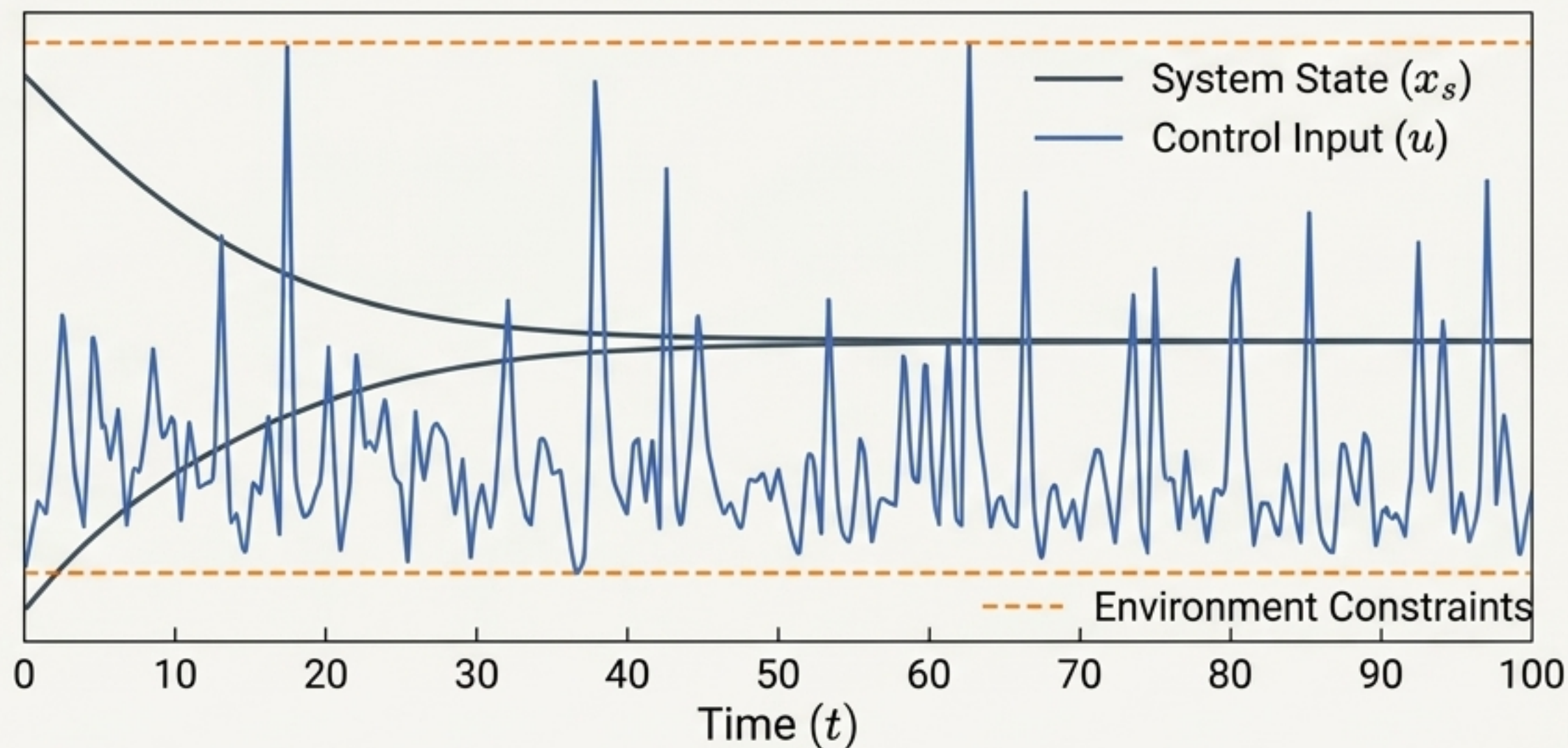
Case Study: Input-Dependent Perception

Setup

Measurement noise strictly decreases as control input u increases.

$$D_e = (1 - \beta[u]_2) \bar{D}$$

The Result



Input-Dependent Perception: The system actively expends control energy solely to improve sensor quality, stabilizing uncertainty without altering its physical destination.

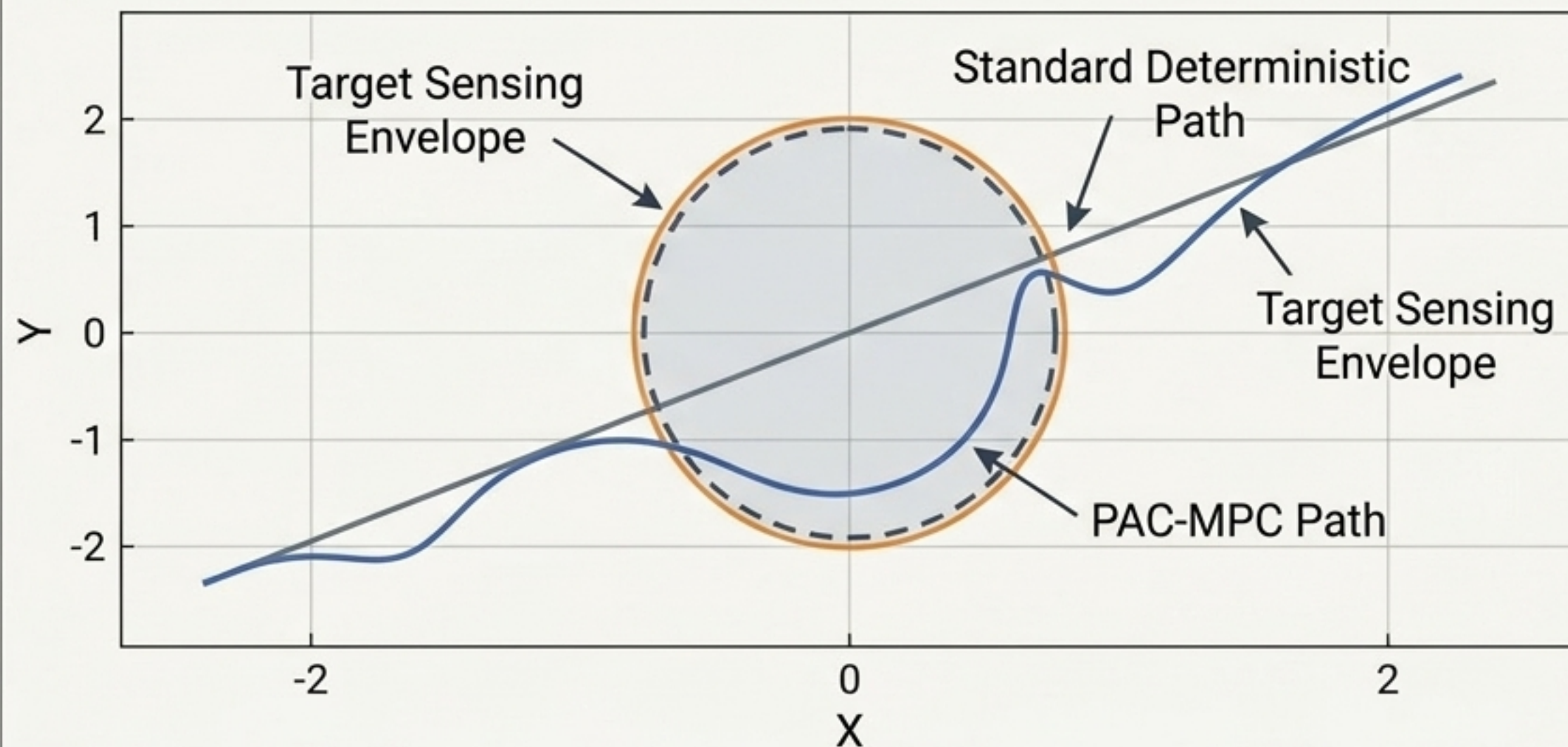
Case Study: State-Dependent Perception

Setup

Measurement quality degrades as distance from target increases.

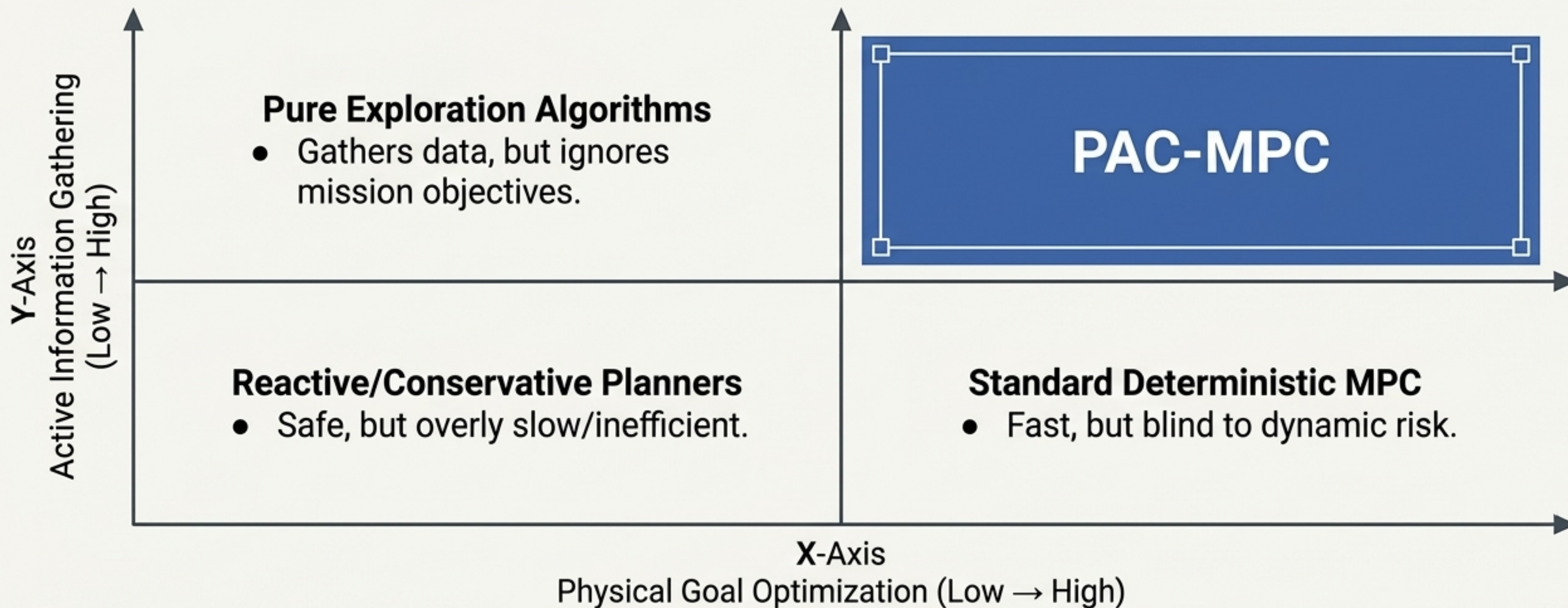
$$[D_e] \propto ([\mu_e] - [x_s])^2$$

The Result



State-Dependent Perception: The system deliberately alters its physical trajectory to prevent field-of-view degradation, taking a longer but fundamentally safer path.

The Synthesis: Surviving the Unknown



By explicitly predicting the interplay between physical action and sensory clarity, PAC-MPC simultaneously optimizes for mission completion and environmental awareness—yielding safe, non-conservative autonomy.